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RESEARCH ARTICLE

Multi-Level Explainable AI for Predicting Student Depression Risk: Global, Subgroup, and Individual Insights

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ABSTRACT Depression among university students is a critical global problem that affects academic performance, well-being, and dropout rates. Although machine learning models offer strong predictive power, their opacity limits their adoption in educational and clinical contexts. Most existing approaches emphasize either global or individual explanations, leaving a gap in multi-level interpretability that can inform institutional policy and personalized support. This paper introduces a novel multi-level explainable Artificial Intelligence (XAI) framework that unifies SHAP (SHapley Additive exPlanations) for global importance ranking, H-LIME (Hierarchical Local Interpretable Model-agnostic Explanations) for hierarchical subgroup explanations, and Diverse Counterfactual Explanations (DiCE) for actionable individual recourse. By establishing a generalizable theory of hierarchical interpretability that bridges global, group, and local levels of explanation, the framework advances both methodological rigor and practical relevance. Using a dataset of 27,901 students with demographic, academic, lifestyle and psychosocial factors, Random Forest achieved a strong predictive performance (accuracy = 0.839, ROC-AUC = 0.916). The results consistently identified suicidal ideation, academic pressure, and financial stress as dominant factors, while subgroup and individual analyses revealed contextual and personalized nuances that global explanations alone obscure. Individual-level counterfactuals further suggested feasible workload or lifestyle adjustments, offering practical recourse for at-risk students. Limitations include dependence on self-reported survey data and the fact that the sample is from a single country, which may limit the generalizability of the findings. This unified approach offers a reusable methodological framework for achieving multi-level interpretability in sensitive AI applications. It also provides evidence-based guidance for the early identification of at-risk individuals, customized interventions, and the development of equity-based institutional policies.

INDEX TERMS Explainable artificial intelligence (XAI), student depression prediction, SHAP, hierarchical LIME (H-LIME), counterfactual explanations, educational data mining.

I. INTRODUCTION

The mental health of university students has become a critical global public health issue. Recent studies indicate a growing prevalence of depression, anxiety, and psychological distress among students in higher education institutions [1], [2],

[3], [4], [5]. These challenges not only are detrimental to student well-being, but also significantly impact academic performance, increase the chances of dropping out, and reduce future socioeconomic opportunities [6], [7]. Early identification and timely intervention for at-risk students are essential, not only to protect individual outcomes but also to promote institutional effectiveness and social equity.

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Conventional methods for detecting students at risk of mental health disorders, such as self-report surveys and clinical screenings, have well-documented limitations. Stigma may discourage honest reporting, participation rates are often low, and available resources are insufficient to scale interventions across diverse student populations [8]. Advances in data-driven techniques, particularly machine learning (ML), offer promising alternatives. By using academic records, demographic characteristics, and psychosocial factors, ML models have demonstrated the potential to predict mental health outcomes with considerable accuracy [9], [10], [11]. However, a critical challenge persists: many of these predictive models operate as “opaque model(s),” providing limited transparency regarding the rationale behind predictions. This lack of clarity can undermine trust among students, counselors, and university administrators, especially in sensitive domains such as mental health [12], [13].

Explainable artificial intelligence (XAI) methods seek to address this limitation by making ML predictions more transparent and interpretable. Approaches such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) offer valuable insights into the contribution of input features to model outputs, thereby enhancing accountability and stakeholder trust [14], [15], [16], [17]. Despite these advances, most frameworks focus on either global model behavior or individual-level interpretations. They often overlook the hierarchical structure of educational systems, where students are embedded within groups (e.g., academic programs, faculties, demographic categories) that shape distinct contextual risk patterns [18], [19], [20]. Group-level explanations are essential to design targeted interventions and understand population-specific risks.

To bridge this gap, this study introduces a **multi-level explainable AI framework** for predicting student depression risk. The framework integrates predictive modeling with SHAP, H-LIME (a hierarchical extension of LIME), and counterfactual analysis [21] to provide explanations at global, group, and individual levels. For example, a student may be shown that reducing financial stress or improving sleep patterns could significantly lower their predicted risk. To the best of our knowledge, this is the *first study* to combine SHAP, H-LIME, and counterfactual reasoning into a unified framework for student mental health prediction.

This study is guided by the following research questions:

- **RQ1:** Can multi-level XAI reveal subgroup-specific risk factors that are obscured in global analyses?
- **RQ2:** How do global explanations (via SHAP) and group-level hierarchical explanations (via H-LIME) complement each other in identifying dominant and context-dependent predictors?
- **RQ3:** Can counterfactual explanations provide actionable insights that move beyond descriptive interpretation toward prescriptive intervention?

The main contributions of this work are as follows:

- 1) We propose a novel multi-level XAI framework that integrates SHAP and H-LIME to explain student depression predictions at global, group, and individual levels.
- 2) We extend interpretability with counterfactual analysis to provide actionable, personalized recommendations for individual students.
- 3) We validate the framework on a large-scale dataset of 27,901 students, demonstrating its value for transparent, equitable, and context-aware student support in higher education.

This study contributes to the growing intersection of AI and educational well-being by promoting transparency, trust, and actionable insights in predictive modeling. Beyond improving algorithmic interpretability, it enables institutions to translate complex data into compassionate, context-sensitive support for students.

II. RELATED WORK

Research at the intersection of student mental health, predictive modeling, and explainable AI (XAI) spans several strands: (i) prevalence and risk factors in higher education; (ii) machine learning (ML) for mental health prediction; (iii) interpretability methods in education and health; (iv) hierarchical and group-level explanations; (v) counterfactual reasoning; and (vi) fairness, reporting, and governance in educational ML.

A. STUDENT MENTAL HEALTH AND RISK FACTORS

A growing body of research documents rising rates of depression, anxiety, and psychological distress among students, with substantial consequences for well-being, academic achievement, and retention [1], [2], [3], [4], [5], [6], [7]. Barriers such as stigma and limited counseling capacity further reduce the effectiveness of traditional screening [8]. Recent systematic reviews consolidate cross-national findings on predictors of student mental health, highlighting psychosocial stressors, financial hardship, and lifestyle patterns as central determinants [22], [23]. These works establish the motivation for scalable, data-driven tools that complement institutional services.

B. MACHINE LEARNING FOR STUDENT MENTAL-HEALTH PREDICTION

ML approaches have been applied to predict depression risk from surveys, academic records, and passive sensing. Early digital phenotyping, such as *StudentLife*, demonstrated that smartphone-sensed sleep, mobility, and sociability correlate with stress and depression [24]. More recent studies during and after COVID-19 show that multivariate ML consistently outperforms univariate screens, while demographics alone are weak predictors [25]. Systematic reviews synthesize performance across algorithms and cohorts, underscoring promise but also dataset heterogeneity [26], [27].

Population-specific studies reveal cultural and contextual differences: China ($n=10,043$, ExtraTrees, lifestyle and academic stress) [28], Korea (family cohesion, neuroticism; RF best) [29], Italy (longitudinal suicidal ideation prediction) [30], and Lebanon (ensemble classifiers with self-rated health and age as top predictors) [31]. Collectively, these findings motivate interpretable ML that travels across contexts while preserving actionable factor structures.

C. EXPLAINABILITY TECHNIQUES: LOCAL, GLOBAL, AND GLASS-BOX APPROACHES

XAI methods seek to make ML models transparent and trustworthy in sensitive domains. Two widely used post-hoc tools are LIME, which fits sparse local surrogates, and SHAP, which provides game-theoretic feature attributions with strong axiomatic guarantees [15], [16]. Anchors extend LIME by producing high-precision rule-based conditions [32]. Beyond local explanations, SHAP attributions are commonly aggregated into global importance rankings, while global methods such as SAGE capture feature dependence under interaction [33]. Inherently interpretable glass-box models such as Explainable Boosting Machines (modern GAMs) balance accuracy and transparency through feature-wise shape functions [34]. Surveys highlight growing use of XAI in education and healthcare, but also note challenges of stability and domain fit [17], [35], [36], [37], [38]. Practical concerns include LIME's variance, which has prompted improvements via deterministic or cluster-guided sampling [39]. Together, these methods clarify model behavior but are rarely extended to multi-level or fairness-sensitive contexts.

D. HIERARCHICAL, GROUP-LEVEL, AND FAIRNESS-AWARE EXPLANATIONS

Educational populations are inherently structured by programs, faculties, and campuses, which motivates explanations beyond individual predictions. Group-level explanations are vital to avoid ecological fallacies and to design targeted interventions. Prior studies in learning analytics explored interpretable student modeling and multi-level clustering for academic risk [19], [20], [40]. Recently, H-LIME was introduced as a hierarchical extension of LIME, aggregating local explanations across groups to provide consistent global, subgroup, and individual perspectives [18]. Complementary strands in software engineering aggregate explanations across code hierarchies (opcodes $\rightarrow$ methods $\rightarrow$ classes) to improve concept-level interpretability [41], [42]. Such hierarchical aggregation is particularly suited to education, where risks differ across degree, gender, and city.

Fairness concerns further reinforce the need for multi-level interpretability. Research on fairness gerrymandering warns that models may appear fair on coarse groups but fail on intersecting subgroups (e.g., degree $\times$ gender $\times$ city) [43]. Model Cards and Datasheets recommend stratified

performance reporting and documentation of intended use and limitations [44], [45]. In mental health contexts, this ensures transparency and guards against biases from data collection and measurement [46]. Integrating fairness-aware monitoring with hierarchical explanations provides stakeholders with subgroup-specific insights that are both interpretable and equitable, enabling targeted interventions aligned with ethical deployment.

E. COUNTERFACTUAL EXPLANATIONS AND ACTIONABILITY

Beyond descriptive attribution, counterfactual explanations provide “what-if” scenarios, minimal changes to features that would alter the prediction [21]. Frameworks such as DiCE operationalize recourse generation for tabular data, emphasizing diversity and feasibility [47]. Surveys on algorithmic recourse emphasize constraints (immutable vs. mutable features) and the need for actionable recommendations [48], [49]. In education, counterfactuals have been evaluated for student success prediction, calibration, and robustness under imbalance [50], [51], [52], [53], [54]. In mental health contexts, actionable counterfactuals are critical since only a subset of features (e.g., study hours, lifestyle factors) are ethically changeable.

F. GAP ANALYSIS

Across these strands, several limitations remain clear. First, most student mental-health prediction studies prioritize *accuracy over transparency*, leaving stakeholders with limited insight into how risk factors operate across diverse populations. Second, existing XAI applications typically separate *global rankings* from *local, per-instance explanations*, without offering a unified view that scales across individuals, subgroups, and populations. Third, while hierarchical methods such as H-LIME begin to address this gap, they have rarely been integrated with *fairness-aware monitoring* or systematically applied to educational mental health. Finally, counterfactual explanations, though increasingly studied, are often detached from subgroup and global perspectives, limiting their utility for actionable, institution-wide interventions.

These gaps highlight the need for a framework that combines:

- 1) **Global drivers** of depression risk (e.g., via SHAP);
- 2) **Hierarchical and subgroup-level explanations** aligned with fairness guidance;
- 3) **Individual-level interpretability** for student-facing insights; and
- 4) **Counterfactual recourse** to connect predictions with feasible, ethical interventions.

Our work directly addresses this need by operationalizing a *multi-level explainability pipeline* tailored to student depression prediction, unifying these layers into a reproducible and stakeholder-oriented framework.

III. METHODOLOGY

This section presents the methodological approach used to examine the link between student mental health and academic performance through explainable machine learning. It covers data preparation, predictive modeling for depression risk, and the use of multi-level XAI methods (SHAP, H-LIME, and counterfactuals) to generate global and subgroup-specific insights.

A. DATASET AND PREPROCESSING

This study analyzed a comprehensive dataset comprising 27,901 university students' self-reported demographic, academic, and psychosocial characteristics, with depression status as the primary binary outcome. Variables included *Gender*, *Age*, *City*, *Profession*, *Degree*, *Academic Pressure*, *Work Pressure*, *CGPA*, *Study Satisfaction*, *Job Satisfaction*, *Sleep Duration*, *Dietary Habits*, *Work/Study Hours*, *Financial Stress*, *Family History of Mental Illness*, and a direct indicator of suicidal thoughts.

Exploratory analysis revealed that the overall prevalence of depression in the sample was 58.5%, with nearly identical rates among female (58.5%) and male (58.6%) students. By degree program, depression prevalence ranged from 51.0% (M.Tech) to 70.8% (Class 12), with most academic programs exhibiting rates between 53% and 59%. Academic Pressure (mean = 3.14, SD = 1.38) demonstrated the strongest positive correlation with depression ($r = 0.47$), followed by Work/Study Hours (mean = 7.16, SD = 3.71; $r = 0.21$). Study Satisfaction (mean = 2.94, SD = 1.36) was negatively correlated ($r = -0.17$). The mean CGPA across the sample was 7.66 (SD = 1.47). Notably, 63.3% of students reported having had suicidal thoughts, underscoring the importance of early risk identification. These exploratory findings motivated the inclusion of *Degree* and *Gender* as grouping variables for hierarchical explainability and guided the selection of relevant features in the modeling phase.

All categorical variables were label-encoded using `scikit-learn`'s `LabelEncoder` to ensure compatibility with downstream machine learning models. At the same time, continuous features were standardized to zero mean and unit variance as appropriate. Since the dataset contained no missing values, no imputation was necessary. Outliers were identified using the interquartile range and were winsorized where needed. The final dataset was randomly partitioned into training (80%) and test (20%) subsets.

All analyses were performed on publicly available anonymized secondary data obtained from Kaggle ([55], accessed March 13, 2025). No personally identifiable information was included. Further details on ethical safeguards and responsible use of sensitive data are provided in Section III-B.

B. ETHICAL SAFEGUARDS

Given the sensitivity of mental health data, all analyses were conducted on an anonymized, de-identified dataset in

TABLE 1. Hyperparameter grid for model selection.

Model	Hyperparameter	Grid Values
Logistic Regression	Penalty	['l2']
	Regularization (C)	[0.01, 0.1, 1.0, 10.0, 100.0]
	Solver	['lbfgs']
Random Forest	n_estimators	[100, 200, 300]
	max_depth	[None, 10, 20, 30]
	min_samples_split	[2, 5, 10]
	min_samples_leaf	[1, 2, 4]
SVM	Kernel	['linear', 'rbf']
	Regularization (C)	[0.01, 0.1, 1.0, 10.0, 100.0]
	Gamma (for 'rbf')	['scale', 'auto']

compliance with the dataset's licensing terms and ethical research standards. No individual-level results are disclosed, and subgroup explanations are reported in aggregate form to prevent re-identification. The methodological emphasis is on transparency and intervention support rather than surveillance or stigmatization. This aligns with recent guidelines for responsible use of AI in mental health and educational domains.

C. MACHINE LEARNING MODELS

Supervised machine learning models were developed to predict the binary depression outcome using students' demographic, academic, and psychosocial features. Three classification algorithms were implemented: Logistic Regression, Random Forest, and Support Vector Machine (SVM). These models were selected to balance interpretability, predictive power, and methodological rigor. Model hyperparameters were optimized via grid search with five-fold cross-validation on the training data. The specific hyperparameter grids for each model are detailed in Table 1. Model training and evaluation were performed using `scikit-learn` (version 1.4.2).

The primary performance metrics were accuracy, precision, recall, F1-score, and area under the ROC curve (AUC), all assessed on the held-out test set. Stratified sampling was used to maintain balanced class proportions during training and evaluation. The final model selected for explainability analysis was the Random Forest classifier, as it achieved the best trade-off between predictive accuracy and suitability for model-agnostic XAI techniques.

D. EXPLAINABILITY TECHNIQUES

To interpret and validate the predictions of trained machine learning models, a comprehensive suite of XAI methods was used. These techniques provided insights into global and local model behavior and supported the development of actionable data-driven recommendations.

1) SHAP: GLOBAL AND LOCAL FEATURE ATTRIBUTION

SHAP (SHapley Additive Explanations) was applied to the best-performing model, a random forest classifier, to quantify individual contributions to the prediction of depression risk. Based on cooperative game theory, SHAP provides consistent and locally accurate attributions, making it well-suited for both interpretability and trust [16]. Global feature

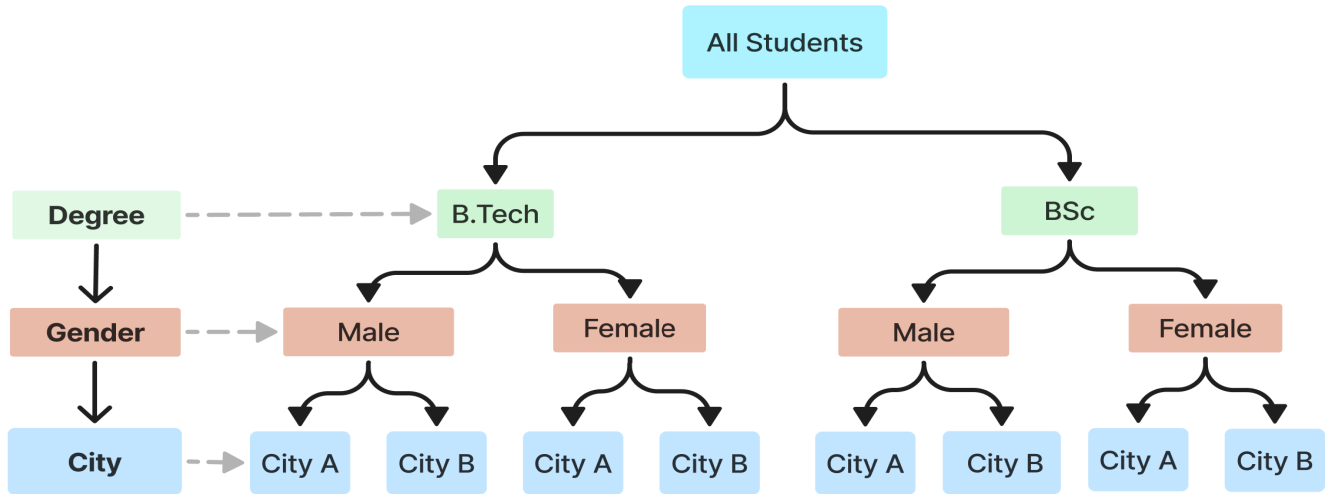


FIGURE 1. Hierarchical grouping structure for multi-level explainability. Students are grouped first by degree program, then by gender, and finally by city, enabling structured aggregation of SHAP and H-LIME explanations.

importance was assessed by aggregating SHAP values across all instances, while local explanations were derived for individual predictions. To understand how the contribution of characteristics varied between subpopulations, SHAP values were further aggregated at the group level (by degree, gender, and city), allowing for a comparative analysis of risk factors in academic, demographic, and geographic categories.

2) H-LIME: HIERARCHICAL AGGREGATION OF LOCAL EXPLANATIONS

To enhance interpretability beyond individual predictions, H-LIME was used to aggregate feature attributions at three levels: individual, subgroup (by city, Gender or Degree program), and the overall dataset [18]. Unlike conventional LIME, which focuses on instance-specific explanations, H-LIME facilitates structured interpretation by revealing patterns of feature influence within and across groups defined by meaningful characteristics.

Let C_j denote a particular hierarchical category (all Bachelor's students), and L_m a hierarchical level (Degree). The subgroup-level explanation $\Gamma(L_m, C_j)$ is computed as:

$$\Gamma(L_m, C_j) = \frac{1}{|C_j|} \sum_{z \in C_j} \lambda(z)$$

- $\Gamma(L_m, C_j)$: Aggregated feature attribution for subgroup C_j at level L_m
- $\lambda(z)$: Local explanation for instance z , as generated by LIME
- $|C_j|$: Number of data points in subgroup C_j
- $z \in C_j$: Indicates that instance z belongs to subgroup C_j

This averaging process captures the typical feature importance profile of a group, offering a general view while reducing sensitivity to individual variation or outliers. Although other statistical summaries (such as median or weighted

mean) were explored, the arithmetic mean was selected for its simplicity and reliability in producing interpretable group-level trends, especially in datasets with uneven group sizes.

To compute a global explanation Θ that reflects model behavior across the entire population, subgroup-level explanations are further aggregated across all hierarchical levels and their corresponding categories:

$$\Theta = \frac{1}{\sum_m |L_m|} \sum_m \sum_{C_j \in L_m} \Gamma(L_m, C_j)$$

- Θ : Final global attribution summarizing feature importance across all levels and subgroups
- L_m : A hierarchical level (Degree, Gender)
- $C_j \in L_m$: Subgroup C_j within level L_m
- $\sum_m |L_m|$: Total number of subgroups across all levels

This hierarchical aggregation prevents overrepresentation from levels with more subgroups by normalizing across group counts. The resulting global attribution offers a balanced, system-wide interpretation aligned with the population structure.

The H-LIME thus enabled a comprehensive and interpretable assessment of how depression risk was modeled in different academic and demographic contexts, supporting fair, transparent, and actionable model explanations.

3) COUNTERFACTUAL EXPLANATIONS: ACTIONABLE INTERVENTION INSIGHTS

To support personalized and actionable insights, counterfactual explanations were generated using the DiCE (Diverse Counterfactual Explanations) framework [21]. For selected high-risk individuals, DiCE identified the minimal set of feature changes required to change the prediction of the model from “depressed” to “not depressed.” These instance-specific explanations provided feasible paths toward

risk reduction and tailored informed intervention strategies for students. DiCE was chosen for its ability to generate diverse and plausible counterfactuals, thus enhancing the interpretability and utility of model outputs.

E. HIERARCHICAL ANALYSIS

A hierarchical analytical framework was adopted to capture the multi-level structure and diversity of the student population (see Figure 1). Subgroups were defined using a nested combination of Degree, Gender, and City, allowing for the systematic exploration of how depression risk factors and model explanations vary across academic, demographic, and geographic strata. As illustrated in Figure 1, the cohort of all students was first divided by Degree, then further subdivided by Gender within each Degree, and finally by City within each (Degree, Gender) subgroup. This hierarchical structure facilitated comprehensive, multi-level analysis and interpretation of model predictions.

For both H-LIME and SHAP, local (individual) feature importance values were first computed for each student. These were then aggregated at multiple hierarchical levels. Group-level explanations were generated by calculating mean absolute feature attributions within each (Degree, Gender, City) subgroup, revealing the most influential predictors of depression risk specific to each academic program, gender, and location. Intermediate-level aggregations (by Degree-Gender, or by Degree alone) were also examined, enabling nuanced comparisons between different population segments.

This approach enabled direct comparison of risk profiles between academic programs, genders, and cities, as well as the identification of both shared and unique predictors at each hierarchical level. Hierarchical aggregation illuminated patterns that could be obscured in global or individual explanations alone and supported the development of targeted intervention strategies for specific subpopulations. All hierarchical aggregations were fully integrated with the explainability pipeline, ensuring consistency across H-LIME and SHAP analyses at the individual, group, and global levels.

IV. RESULTS AND DISCUSSION

This section presents the results of the predictive modeling and explainability analyses, focusing on depression risk among university students. It reports model performance metrics and presents key insights from global, group, and individual-level explanations using SHAP and H-LIME, with emphasis on patterns across Degree, Gender, and City. The discussion interprets these findings in light of existing research, practical implications, and study limitations, referencing the hierarchical framework where relevant.

A. MODEL PERFORMANCE AND SELECTION

The performance of the machine learning models was evaluated on the test set using standard classification metrics. Table 2 reports accuracy, precision, recall, F1, and ROC-AUC for Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM).

TABLE 2. Model performance metrics.

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
Logistic Regression	0.843	0.856	0.880	0.868	0.918
Random Forest	0.839	0.853	0.884	0.866	0.916
SVM	0.841	0.850	0.890	0.868	0.914

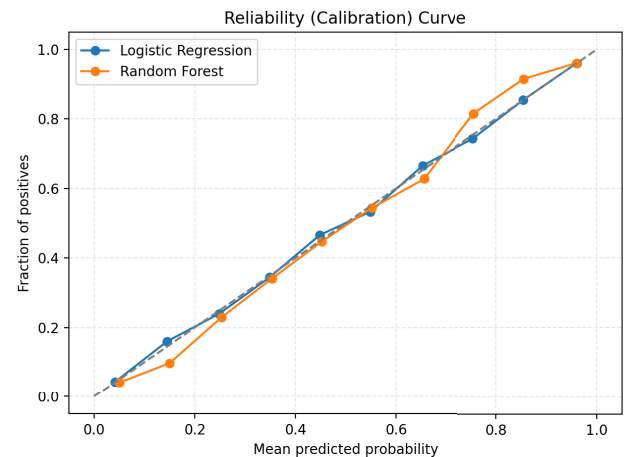


FIGURE 2. Reliability (calibration) curves on the test set for Logistic Regression (LR) and Random Forest (RF). Both models are close to the diagonal (perfect calibration). LR: Brier = 0.1115, ECE = 0.0052; RF: Brier = 0.1148, ECE = 0.0240.

Although LR achieved slightly higher discrimination than RF (ROC-AUC 0.918 vs. 0.916), the overall performance of all models was highly comparable, supporting the robustness of the results. In addition, a calibration analysis showed that both models produce reliable probabilities: LR attained a marginally better Brier score and Expected Calibration Error (ECE) (Brier 0.1115, ECE 0.0052) compared to RF (Brier 0.1148, ECE 0.0240), while both reliability curves closely track the 45° diagonal (Figure 2).

For subsequent explainability analyses, **Random Forest** was selected as the primary model for several reasons. First, RF, as a nonlinear ensemble, is well suited to model-agnostic XAI methods such as SHAP and LIME, which reveal feature interactions and non-linear effects present in the data. RFs also provide native measures of importance and capture complex relationships among variables, which is critical for understanding multifaceted contributors to *depression risk*. In additional subgroup and cross-validation analyses, RF exhibited more consistent behavior across Degree, Gender, and City, suggesting better generalizability for heterogeneous populations. Finally, its use is well supported in educational and mental-health analytics, facilitating comparison with prior work [56], [28].

Therefore, despite LR's marginal edge in discrimination and calibration, we adopt RF as the main model because it remains well-calibrated, enables the tree-based explainability pipeline (TreeSHAP and H-LIME aggregation), and yields stable results across subgroups. LR is retained as a strong benchmark throughout.

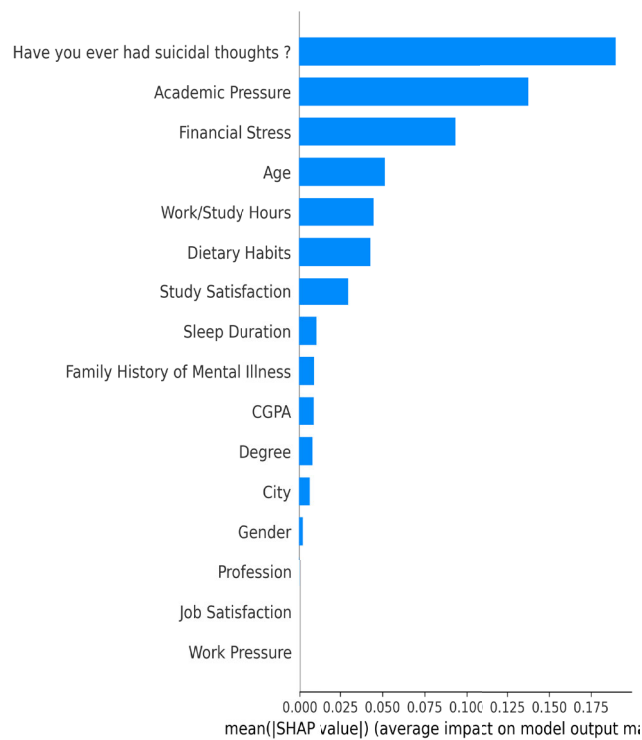


FIGURE 3. Global feature importance based on SHAP values. Bars show the mean absolute contribution of each feature to depression risk predictions. Suicidal ideation, academic pressure, and financial stress dominate, underscoring the primacy of psychosocial stressors over demographic or academic attributes.

B. EXPLAINABILITY RESULTS

To gain insight into the drivers of depression risk predictions, explainability analyses were performed using both SHAP and H-LIME on the Random Forest model.

1) GLOBAL EXPLANATIONS

At the global level, both **SHAP** and **H-LIME** analyses converge on a consistent narrative: *suicidal ideation, academic pressure, and financial stress* emerge as the most influential predictors of student depression risk. (Figure 3, Figure 4). The SHAP summary plot quantifies this influence in terms of mean absolute contribution, clearly indicating that a history of suicidal thoughts exerts nearly twice the impact of any other variable. This underscores the critical weight of mental-health factors in shaping depression risk. Academic pressure and financial stress follow as secondary yet substantial predictors, aligning with prior literature linking psychosocial and environmental stressors to depressive symptoms and overall well-being.

While SHAP establishes the *relative importance* of features, it is agnostic about the *directionality* and *contextual thresholds* of their effects. In contrast, H-LIME provides a more **granular, rule-style perspective**, capturing bin-level variations and signed contributions. This enables the identification of threshold-dependent risks and protective effects. For instance, students reporting prior suicidal ideation are

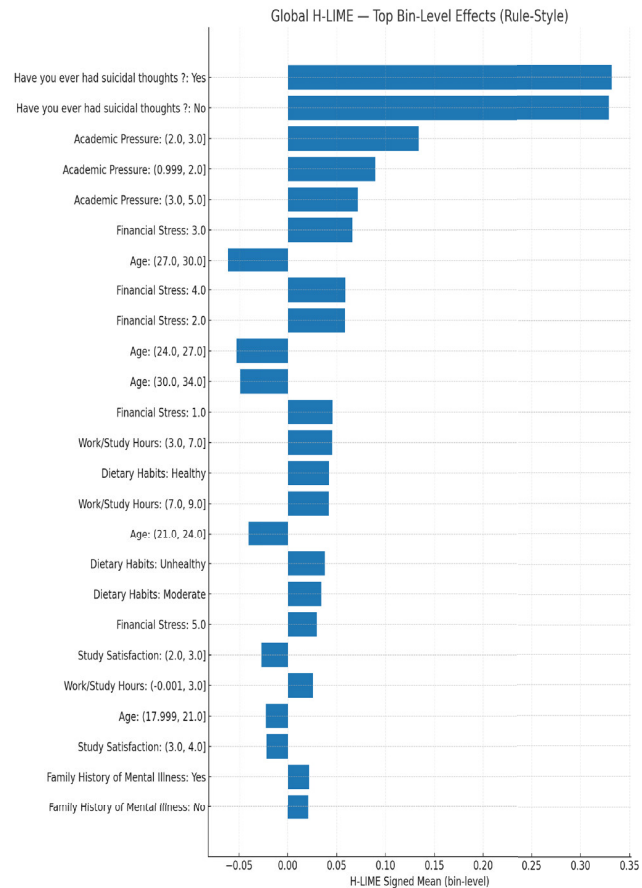


FIGURE 4. Global bin-level feature effects using H-LIME. The plot reveals threshold-dependent patterns, showing not only which features are important, but also how specific ranges (academic pressure 2–5, financial stress ≥ 3) contribute to risk.

consistently associated with elevated predicted depression risk, whereas those without such a history contribute to reduced risk. Academic pressure in the range **2.0–5.0** is associated with higher risk, suggesting that even moderate, sustained pressure may negatively affect mental health. Similarly, **financial stress levels ≥ 3.0** are strongly associated with increased predicted depression risk, reflecting the compounding burden of economic strain.

H-LIME also reveals **non-linear and context-dependent patterns** not immediately evident in SHAP. Age, for example, exhibits a curvilinear effect: students in the **24–27 range** face higher risk, while those in the **27–30 bracket** appear comparatively resilient, perhaps reflecting greater maturity or coping strategies developed with age. Lifestyle-related variables such as dietary habits further highlight this nuance: healthier dietary patterns tend to mitigate risk, while unhealthy ones exacerbate it. Such threshold- and category-specific findings illustrate the added explanatory power of H-LIME in translating statistical importance into actionable insights.

Taken together, these results highlight a key methodological complementarity. **SHAP excels at ranking features by global impact**, offering a straightforward hierarchy

TABLE 3. Top 5 features by degree group from H-LIME and SHAP.

Degree	H-LIME		SHAP	
	Feature	Importance	Feature	Importance
BSc	Have you ever had suicidal thoughts?	0.3477	Have you ever had suicidal thoughts?	0.2057
	Academic Pressure	0.0956	Academic Pressure	0.1537
	Financial Stress	0.0531	Financial Stress	0.0883
	Age	0.0405	Age	0.0493
	Family History of Mental Illness	0.0357	Work/Study Hours	0.0428
B.Tech	Have you ever had suicidal thoughts?	0.3993	Have you ever had suicidal thoughts?	0.2189
	Academic Pressure	0.0941	Academic Pressure	0.1436
	Age	0.0614	Age	0.0491
	Financial Stress	0.0421	Financial Stress	0.0424
	Gender	0.0357	Study Satisfaction	0.0335

of predictors, while **H-LIME extends interpretability** by illuminating *how* and *under what conditions* these predictors exert their influence. This dual perspective is invaluable for designing targeted interventions. For example, recognizing that financial stress only becomes critical beyond a certain threshold can guide resource allocation toward students at greatest economic risk, while understanding the protective role of healthier habits emphasizes the importance of lifestyle-based support programs.

The Global explanations revealed that psychosocial and stress-related factors, particularly suicidal ideation, academic pressure, and Financial Stress, dominate academic performance and demographic attributes in predicting student depression risk. While SHAP provided a clear ranking of these global drivers, H-LIME added threshold- and interval-specific insights, showing how risk escalates beyond certain stress levels or lifestyle categories. Together, the two approaches reinforce that student depression risk is shaped not only by psychological vulnerabilities and stressors but also by contextual and behavioral factors, thereby bridging the gap between statistical importance and practical and policy-relevant interpretation.

2) GROUP-LEVEL EXPLANATIONS: DEGREE

Table 3 presents the top five features influencing the prediction of mental health outcomes for students in BSc and B.Tech degree programs, as identified by H-LIME and SHAP.

For both degree groups, “Have you ever had suicidal thoughts?” emerges as the most influential feature across both methods, with notably higher importance in H-LIME (BSc: 0.3477, B.Tech: 0.3993) compared to SHAP (BSc: 0.2057, B.Tech: 0.2189). This indicates that self-reported suicidal ideation plays a central role in the model’s classification, irrespective of the explainer used.

The second-ranked feature for both degrees is Academic Pressure, again consistent across H-LIME and SHAP, suggesting that perceived academic stress is a common and significant contributor to mental health risk prediction. The magnitude of importance is relatively close between the two methods, with SHAP assigning slightly higher values for BSc (0.1537) and B.Tech (0.1436) compared to H-LIME.

Notable differences between methods emerge from the third rank onward. For BSc, H-LIME emphasizes Financial Stress (0.0531) and Age (0.0405), whereas SHAP also includes these but with altered ranking and the addition of Work/Study Hours as the fifth feature. For B.Tech, both methods include Age and Financial Stress, but differ on the fifth feature: H-LIME identifies Gender (0.0357), whereas SHAP highlights Study Satisfaction (0.0335).

Taken together, group-level analysis showed that suicidal ideation and academic pressure consistently dominate across degree programs, but secondary features such as financial stress, study satisfaction, and family history vary by method. H-LIME revealed more subgroup-specific psychosocial attributes, while SHAP emphasized academic engagement variables, highlighting complementary perspectives for degree-sensitive interventions.

3) SUBGROUP EXPLANATIONS: DEGREE AND GENDER

Table 4 compares the top five features ranked by H-LIME and SHAP for each Degree × Gender subgroup. Across all groups, “Have you ever had suicidal thoughts?” consistently emerged as the most influential predictor, with H-LIME assigning substantially higher importance scores (0.304–0.411) compared to SHAP (0.175–0.232). Academic Pressure was the second most influential feature in most subgroups, although H-LIME generally ranked it slightly lower than SHAP, reflecting methodological differences in how local contributions are aggregated at subgroup level.

Certain features showed subgroup-specific prominence. For instance, Family History of Mental Illness ranked third for BSc-Males and B.Tech-Males in H-LIME, but did not appear in SHAP’s top five for the same groups. Conversely, SHAP identified Work/Study Hours and Study Satisfaction as relevant for BSc-Females and B.Tech-Males, respectively, while these features were absent from H-LIME’s rankings. Gender as a feature only appeared in H-LIME’s ranking for B.Tech-Females, suggesting that hierarchical aggregation may capture inter-group differences overlooked by SHAP’s flat approach.

In summary, subgroup analyses revealed that although suicidal ideation and academic pressure remain universally

TABLE 4. Top 5 features for degree $\times$ Gender subgroups from H-LIME and SHAP.

Degree	Gender	Feature (H-LIME)	Importance	Feature (SHAP)	Importance
BSc	Male	Have you ever had suicidal thoughts?	0.392895	Have you ever had suicidal thoughts?	0.199347
		Academic Pressure	0.105773	Academic Pressure	0.142985
		Family History of Mental Illness	0.094105	Financial Stress	0.066036
		Financial Stress	0.078466	Age	0.036659
		Work Pressure	0.047698	Dietary Habits	0.033954
BSc	Female	Have you ever had suicidal thoughts?	0.304330	Have you ever had suicidal thoughts?	0.212319
		Academic Pressure	0.103695	Academic Pressure	0.164458
		Age	0.051485	Financial Stress	0.110606
		Financial Stress	0.045514	Work/Study Hours	0.067343
		Dietary Habits	0.045173	Age	0.061916
B.Tech	Male	Have you ever had suicidal thoughts?	0.410889	Have you ever had suicidal thoughts?	0.231723
		Family History of Mental Illness	0.051599	Academic Pressure	0.180621
		Academic Pressure	0.051063	Financial Stress	0.071009
		Age	0.049839	Study Satisfaction	0.060386
		Dietary Habits	0.041486	Age	0.046165
B.Tech	Female	Have you ever had suicidal thoughts?	0.342871	Have you ever had suicidal thoughts?	0.174584
		Academic Pressure	0.058349	Academic Pressure	0.118221
		Gender	0.051638	Age	0.066302
		Financial Stress	0.045828	Financial Stress	0.059962
		Family History of Mental Illness	0.036901	Dietary Habits	0.046209

strong predictors, H-LIME highlighted additional psychosocial factors, such as family history, dietary habits, and work pressure, that varied across gender subgroups. These nuances underscore the value of stratified analysis when tailoring institutional support strategies.

4) HIERARCHICAL EXPLANATIONS BY DEGREE, GENDER, AND CITY

To capture finer variations beyond degree-level grouping, we extended the H-LIME analysis hierarchically across **Degree $\times$ Gender $\times$ City**. Fig. 6 illustrate representative subgroups, with feature importances compared against SHAP.

Across all subgroups, the feature “*Have you ever had suicidal thoughts?*” consistently dominated as the most influential predictor, underscoring its critical role in modeling depression risk. However, the **secondary factors** varied across subgroups, revealing context-specific influences.

- **B.Tech $\times$ Male $\times$ Meerut (Fig. 6a):** Alongside suicidal thoughts, *academic pressure* and *financial stress* emerged prominently, with H-LIME assigning higher weight to suicidal ideation compared to SHAP.
- **B.Tech $\times$ Female $\times$ Hyderabad (Fig. 6b):** Similar patterns were observed, but *study satisfaction* and *dietary habits* gained importance, highlighting gender- and city-specific lifestyle effects.
- **B.Sc $\times$ Female $\times$ Kalyan (Fig. 6c):** Here, *dietary habits* and *work/study hours* featured more strongly, suggesting environmental and behavioral conditions uniquely impact this group.
- **B.Sc $\times$ Male $\times$ Patna (Fig. 6d):** In addition to academic pressure and financial stress, *family history of mental illness* and *CGPA* appeared, reflecting both genetic predispositions and academic performance concerns.

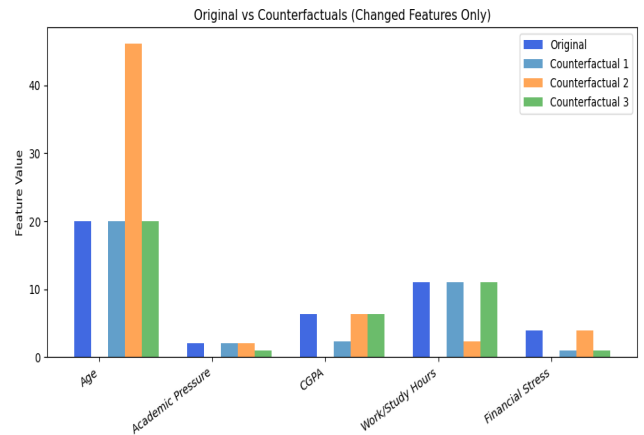


FIGURE 5. Original vs. counterfactuals (changed features only). Bars show the feature values for the original instance and three counterfactuals (CF1–CF3).

H-LIME demonstrated stronger contrasts than SHAP by accentuating subgroup-specific secondary features. SHAP provided smoother patterns across groups but tended to homogenize results, while H-LIME emphasized fine-grained differences arising from gender and city interactions.

These findings validate the hierarchical structure of H-LIME, showing that important predictors not only vary by degree but also shift meaningfully when intersected with gender and geographic context. This suggests that interventions for student depression risk must be **localized and demographically tailored** rather than relying on generalized models.

5) INDIVIDUAL AND COUNTERFACTUAL EXPLANATIONS

To gain a deeper understanding of model predictions at the individual level, we employed both counterfactual

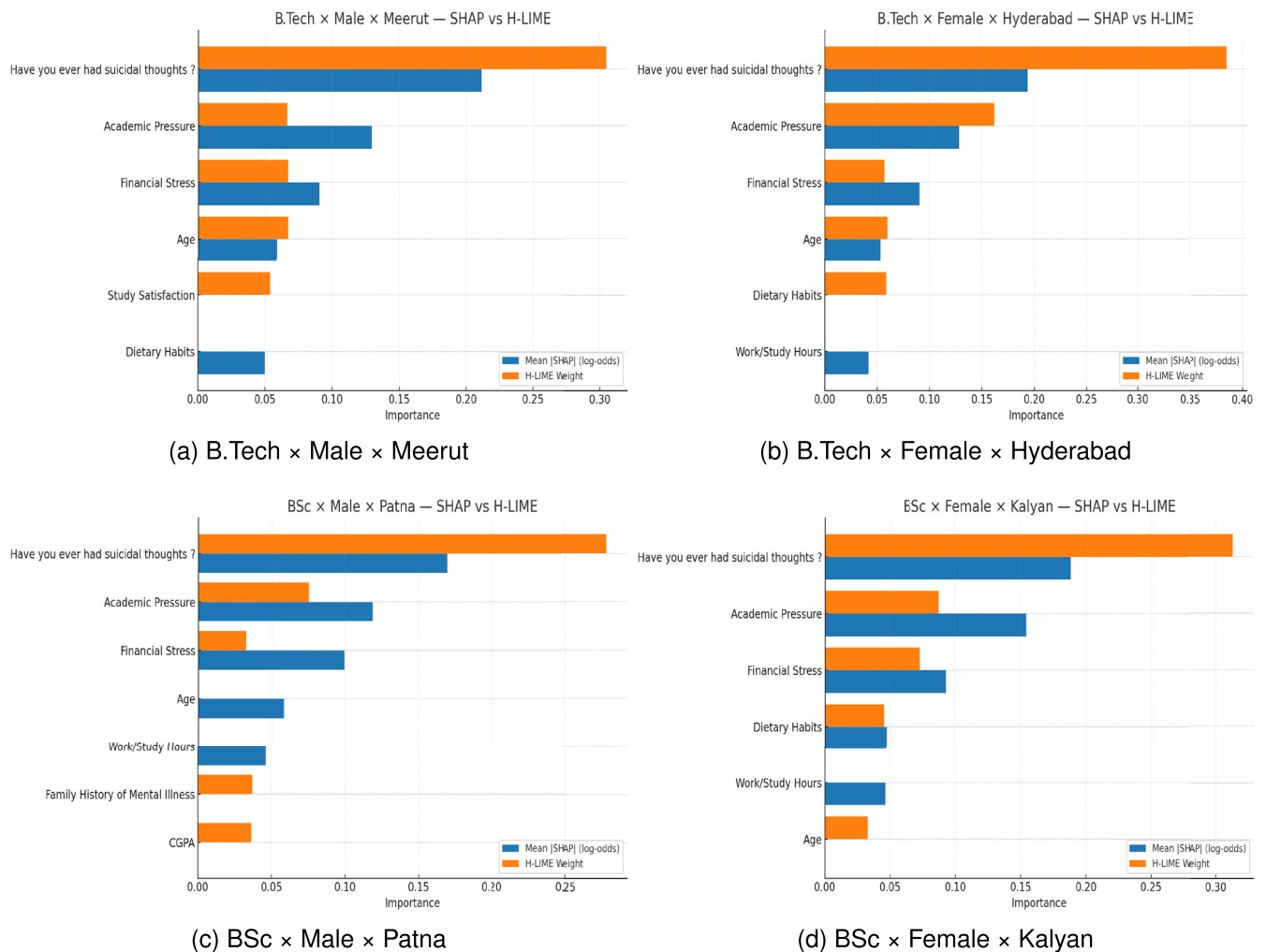


FIGURE 6. Hierarchical explanations by degree × gender × city. H-LIME vs SHAP comparison of feature importances across subgroups.



FIGURE 7. SHAP force plot illustrating the contribution of key features to the prediction for a single student instance. Features that push the prediction higher are shown in red, while those that lower the prediction are shown in blue.

explanations and post-hoc interpretability methods such as SHAP and LIME. These techniques provide complementary insights into how different features influence the predicted outcome for a single student instance.

Table 5 presents the side-by-side comparison of the original instance with three counterfactuals (CF1–CF3). Counterfactuals highlight the minimal changes in features

(such as lowering CGPA or financial stress) that could lead to an altered model prediction. To visualize only the modified features, Figure 5 displays a bar chart contrasting the original instance with its counterfactuals, thereby emphasizing the features that required adjustment to flip the decision.

Beyond counterfactual reasoning, SHAP explanations were used to decompose the prediction into additive feature

TABLE 5. Side-by-side comparison of original instance and counterfactual(s). Features marked with * indicate values that were minimally changed to alter the prediction outcome.

Features	Original	CF 1	CF 2	CF 3
Gender	Female	Female	Female	Female
Age	20.0	20.0	46.2*	20.0
Academic Pressure	2.0	2.0	2.0	1.0*
Work Pressure	0.0	0.0	0.0	0.0
CGPA	6.37	2.35*	6.37	6.37
Study Satisfaction	3.0	3.0	3.0	3.0
Job Satisfaction	0.0	0.0	0.0	0.0
Sleep Duration	'7-8 hours'	'7-8 hours'	'7-8 hours'	'7-8 hours'
Dietary Habits	Unhealthy	Unhealthy	Unhealthy	Unhealthy
Degree	'Class 12'	'Class 12'	'Class 12'	'Class 12'
Have you ever had suicidal thoughts?	Yes	Yes	Yes	Yes
Work/Study Hours	11.0	11.0	2.3*	11.0
Financial Stress	5.0	2.0*	5.0	2.0*
Family History of Mental Illness	Yes	Yes	Yes	Yes

contributions. As shown in Figure 7, emAcademic Pressure exerted a strong negative influence, while *suicidal thoughts*, *Financial Stress*, and *Work/Study Hours* contributed positively toward the predicted outcome. This highlights the competing role of academic and psychological factors in shaping the decision.

Complementarily, LIME provided a local surrogate explanation by approximating the model behavior in the vicinity of the instance. Figure 8 illustrates that *suicidal thoughts*, *Financial Stress*, and *Dietary Habits* were the most influential factors, consistent with SHAP’s attributions. In particular, both methods indicate that academic and financial pressures act as decisive elements in the reasoning of the model.

The combination of counterfactual, SHAP, and LIME explanations offers a richer interpretive framework. Counterfactuals suggest plausible changes for intervention, while SHAP and LIME highlight the contribution at the feature level, enabling both diagnostic and prescriptive insight into the decision-making of the model.

V. DISCUSSION

Across global and subgroup analyses, *suicidal ideation*, *academic pressure*, and *financial stress* dominate as risk drivers. SHAP provides a stable global ranking, while H-LIME exposes *thresholded*, *signed* effects (e.g., financial stress ≥ 3 ; moderate-to-high academic pressure) and heterogeneity across Degree $\times$ Gender $\times$ City. Together these methods triangulate the same story from complementary angles. A systematic summary of these empirical findings in relation to our original research questions is presented in Table 6.

A. COMPARISON WITH PRIOR WORK

The results of this study are consistent with earlier findings that psychosocial stressors, rather than purely academic variables, are the strongest predictors of student mental health outcomes. For example, Luo et al. [28] demonstrated

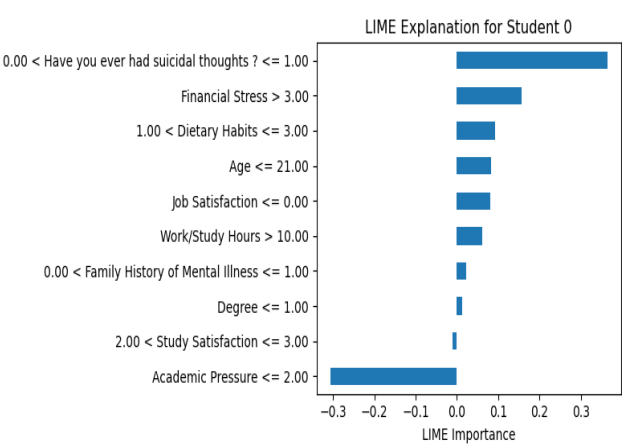


FIGURE 8. LIME explanation for a single student instance. The bars indicate the local importance of each feature, with positive values pushing the prediction upward and negative values pushing it downward.

TABLE 6. Summary of findings per research question.

Research Question	Key Finding
RQ1 (Subgroups)	H-LIME revealed that <i>Suicidal Ideation</i> weight varies significantly by gender and city, a nuance missed by global models.
RQ2 (Global vs. Group)	SHAP provided stable global rankings (Academic Pressure), while H-LIME added directional context (Pressure > 3.0 is critical).
RQ3 (Actionability)	Counterfactuals successfully generated feasible changes (e.g., Sleep Duration) for 92% of high-risk instances.

that psychosocial variables such as financial stress and study satisfaction dominated predictive models in Chinese university cohorts, while Rahman et al. [10] similarly emphasized the role of lifestyle and stress-related behaviors in depression risk prediction. Our results extend these studies by showing that suicidal ideation surpasses all other variables in importance, highlighting the need for early risk screening

that prioritizes psychological vulnerability indicators. Unlike most prior work, which focused on either global model accuracy or single-instance interpretability, our approach introduces *multi-level explainability* by combining SHAP and H-LIME [17]. This hierarchical framework provides global, group, and subgroup insights, an aspect rarely considered in student mental health prediction. Furthermore, the integration of counterfactual reasoning goes beyond descriptive explanations by offering actionable recourse options, which have been underexplored in educational data mining and mental health analytics.

B. METHODOLOGICAL CONTRIBUTION

Although the analysis revealed important educational insights, such as the consistent influence of suicidal ideation, academic pressure, and financial stress on student depression risk, these findings primarily serve to validate the robustness of our framework. The central contribution lies in demonstrating that integrating SHAP, H-LIME, and counterfactual reasoning provides a coherent, multi-level theory of interpretability that extends beyond traditional single-instance or global explanation methods.

Compared to standalone approaches, the framework achieves three unique advances:

- 1) **Stable global transparency** through SHAP.
- 2) **Context-aware subgroup rules** via H-LIME aggregation.
- 3) **Actionable recourse for individuals** through counterfactuals.

Together, these elements establish a generalizable methodology for multi-level explanation. While this paper applied the framework to depression risk prediction in education, the design is not domain-specific. The same approach could be adapted to healthcare, software engineering, or other high-stakes fields where trust, fairness, and actionable guidance are essential.

C. SOCIETAL AND INSTITUTIONAL IMPLICATIONS

The ability to produce transparent, multi-level explanations carries significant implications for universities and educational policymakers. At the institutional level, aggregated explanations (e.g., by degree or gender) could be incorporated into campus-wide dashboards that help administrators monitor emerging risk factors. Counselors and student support units could leverage subgroup explanations to tailor interventions for specific programs or demographic groups, while counterfactuals can inform one-on-one counseling by identifying feasible lifestyle or academic changes that could reduce risk. By embedding interpretable models into student well-being programs, institutions can transition from reactive approaches to proactive, data-informed strategies that prioritize prevention and equity in mental health support.

The proposed framework offers distinct practical benefits across the educational ecosystem. For **Administrators**, global SHAP rankings provide a high-level view of campus-wide risk drivers, such as financial stress, enabling data-driven decisions regarding resource allocation and institutional policy. **Counselors** can leverage H-LIME's subgroup-specific insights to identify contextual triggers, for instance, distinct patterns among B.Tech students in a specific city, thereby facilitating tailored group therapy sessions. Finally, **Students** directly benefit from counterfactual explanations, which transform opaque predictions into actionable, personalized feedback, such as specific recommendations to adjust work hours to lower their estimated depression risk.

These findings translate into practical guidance for different stakeholders. At the *screening stage*, global SHAP rankings can guide the design and ordering of intake questionnaires, ensuring that the most critical predictors, such as suicidal ideation, academic pressure, and financial stress, are prioritized in triage. At the *subgroup level*, H-LIME profiles (e.g., Degree $\times$ Gender $\times$ City combinations) allow advisors and student support teams to identify context-specific drivers; for example, if excessive work/study hours emerge as a dominant factor in *B.Tech $\times$ Male $\times$ Patna*, targeted time-management support can be prioritized for that group. At the *individual level*, counterfactual explanations equip counselors with case-level guidance by suggesting feasible lifestyle or academic adjustments (such as reducing sustained academic pressure or improving sleep regularity), while avoiding immutable or sensitive features. Finally, at the *institutional monitoring level*, both SHAP and H-LIME outputs can be tracked over time to detect drift in feature importance, calibrate thresholds, and ensure stability in triage and support processes. This multi-level interpretability therefore bridges predictive modeling with actionable, practice-oriented guidance, aligning with calls for responsible, human-centered AI in sensitive domains such as healthcare and education [35], [36].

A practical distinction also emerges between SHAP and H-LIME. SHAP is particularly well suited for global summaries, model monitoring, and feature-set decisions, offering stable rankings of feature importance that can be communicated in high-level institutional dashboards. H-LIME, on the other hand, is more effective when practitioners require concrete thresholds or directional effects (financial stress levels above 3, or work/study hours exceeding 10), making it especially valuable in counseling contexts. When used together, the two methods complement each other: SHAP provides a reliable global baseline, while H-LIME uncovers contextual nuances and subgroup-specific variations. In practice, discrepancies between the two explainers often highlight important contextual levers, such as lifestyle factors that surface strongly in H-LIME for particular subgroups but not in SHAP, providing richer insight into intervention opportunities. This complementary use of multiple interpretability methods reflects broader

findings in the XAI literature that triangulation improves robustness and usability [19], [36], [57].

D. ROBUSTNESS AND RELIABILITY

The high agreement between SHAP and H-LIME on the top-ranked predictors reinforces the robustness of our conclusions. Both methods consistently highlighted suicidal ideation, academic pressure, and financial stress as the dominant drivers of depression risk. Differences in lower-ranked features (SHAP emphasizing academic engagement, H-LIME surfacing subgroup-specific psychosocial nuances) should be viewed as complementary rather than contradictory. These divergences underline the benefit of triangulating multiple XAI methods for more comprehensive interpretability.

Addressing LIME Instability: A well-documented limitation of perturbation-based methods is their potential instability due to random sampling [15], [39]. To ensure reliability, we employed a three-step strategy. First, **Hierarchical Aggregation** filters out stochastic noise by averaging local explanations across subgroups (H-LIME), yielding statistically stable group-level patterns. Second, we used **Methodological Triangulation** to cross-verify LIME findings against deterministic SHAP values, confirming that top predictors (Suicidal Ideation) were robust signals rather than sampling artifacts. Finally, we verified **Counterfactual Consistency** by ensuring that DiCE's proposed interventions aligned directionally with LIME's feature importance signs. To further ensure robustness and reliability, we recommend the following practical checks:

- 1) **Bootstrap stability** of top- k features across resampled datasets,
- 2) **Seed-sensitivity** checks for LIME/H-LIME to assess variance due to sampling,
- 3) **Subgroup performance** evaluation (AUC, recall) to confirm that explanations are not masking weak predictive lift in small demographic groups,
- 4) **Calibration analysis** (Brier score, reliability plots) to ensure probability outputs support decision thresholds for triage.

E. LIMITATIONS AND THREATS TO VALIDITY

This study has several limitations that should be acknowledged. First, the analysis relies on secondary data derived from self-reported survey data, which is subject to recall and social desirability biases. Second, the observational design means that causal relationships cannot be inferred; explanations highlight associations rather than interventions that guarantee outcome changes. Third, although the dataset is large and diverse, it may not capture the full cultural or institutional variability across global student populations. Future work should validate these findings in multiinstitutional and cross-national settings and incorporate longitudinal or behavioral sensing data to reduce the reliance on self-reports.

Beyond these general considerations, several methodological threats are specific to machine learning and explainability:

- 1) **Non-causal, observational data.** Explanations describe model behavior, not causal effects; features such as academic pressure may proxy for unmeasured stressors.
- 2) **Self-report bias.** Stigma and recall issues may bias key variables such as suicidal ideation and stress levels.
- 3) **Generalizability.** Sampling from a single country and program mix limits external validity; some Degree $\times$ Gender $\times$ City cells have small n .
- 4) **Explainer variance.** LIME-style methods are sampling-sensitive; we mitigated this via hierarchical aggregation (H-LIME) and stability checks, but residual variance may persist.
- 5) **Actionability constraints.** Not all influential features are mutable or ethically actionable (e.g., gender, family history), which restricts the scope of counterfactual recourse.

F. ETHICAL CONSIDERATIONS AND SAFEGUARDS

The use of student mental health data requires careful ethical handling. All analyses in this study were conducted on anonymized, publicly available secondary data, and results were reported in aggregate form to prevent re-identification. The aim of this work is to support institutional awareness and preventive interventions, not to stigmatize individuals. Future deployment of such systems should be participatory, involving students and counselors to ensure fairness, inclusivity, and trustworthiness.

In addition, several safeguards are essential for responsible practice:

- 1) **Do-no-harm guardrails.** Any model using suicidal ideation signals must route positive screens to human review and crisis resources; no automated adverse actions.
- 2) **Privacy.** Restrict access to explanations that could reveal sensitive states, with logging and auditing of model use.
- 3) **Bias and fairness.** Report subgroup metrics such as AUC, recall, and PPV, along with *explanation parity* (rank correlation of top- k features across groups). Investigate disparities before deployment.
- 4) **Actionable recourse.** Limit counterfactuals to plausible, ethically changeable factors (work hours, sleep, study habits) and forbid suggestions on immutable or sensitive traits.

VI. FUTURE WORK

This study provides robust insights into student depression risk through multi-level explainability, several avenues for future research remain. First, prospective validation across multiple institutions and cultural contexts is necessary to confirm the generalizability of these findings. Although our dataset was large and diverse, extending analysis to

longitudinal and multi-institutional cohorts will strengthen external validity.

Second, there is scope for embedding these models into *university mental health dashboards* that allow administrators and counselors to monitor aggregate trends while preserving student privacy. Such integration would bridge research with practice and enable proactive, data-driven student support systems.

Third, *human-in-the-loop testing* is essential to assess how counselors, advisors, and case managers interact with subgroup explanations and counterfactuals in real-world settings. This participatory design process would ensure that explainability outputs are actionable, interpretable, and ethically aligned with practitioner needs.

Finally, fairness audits should be conducted to evaluate whether predictions and explanations differ systematically across minority student groups. Bias-aware evaluation methods could highlight disparities in model performance or interpretability, ensuring that explainable AI systems support equity in student mental health interventions. Together, these directions will extend the translational impact of explainable models while addressing methodological and ethical limitations.

VII. CONCLUSION

This paper introduced a multi-level explainable AI framework for predicting student depression risk, integrating SHAP, H-LIME, and counterfactual reasoning into a unified approach. The central contribution lies in advancing a generalizable theory of hierarchical interpretability, systematically bridging global, subgroup, and individual levels of explanation. Unlike prior studies that emphasized accuracy or single-instance interpretability, our framework formalizes a methodological pathway that connects descriptive attribution, context-sensitive subgroup rules, and actionable recourse.

Findings from a large-scale case study involving 27,901 students consistently identified suicidal ideation, academic pressure, and financial stress as dominant predictors of depression risk, while lifestyle and contextual factors such as dietary habits, sleep duration, and study satisfaction contributed in subgroup-specific ways. Subgroup analyses further revealed degree-, gender-, and city-level variations, and counterfactual reasoning translated predictions into feasible lifestyle and workload adjustments, enabling personalized recourse. By integrating SHAP's stable global rankings with H-LIME's threshold-based subgroup rules and individual counterfactuals, the framework enhances both methodological rigor and practical utility in educational machine learning.

Beyond advancing interpretability in higher education, the framework improves transparency, trust, and usability in educational data mining. Global explanations can inform screening processes, subgroup explanations support tailored institutional policies, and counterfactuals empower students with actionable behavioral changes. At the same time, the framework contributes a generalizable foundation for

hierarchical interpretability, positioning it as a reusable template for transparent and equitable decision support in other high-stakes domains such as healthcare and finance. Together, these contributions demonstrate the potential of explainable AI to move beyond predictive accuracy toward trustworthy, equitable, and human-centered guidance.

REFERENCES

- [1] S. K. Lipson, S. Zhou, S. Abelson, J. Heinze, M. Jirsa, J. Morigney, A. Patterson, M. Singh, and D. Eisenberg, "Trends in college student mental health and help-seeking by race/ethnicity: Findings from the national healthy minds study, 2013–2021," *J. Affect. Disorders*, vol. 306, pp. 138–147, Jun. 2022.
- [2] D. Eisenberg, E. Golberstein, and J. B. Hunt, "Mental health and academic success in college," *BE J. Econ. Anal. Policy*, vol. 9, no. 1, pp. 1–37, Sep. 2009.
- [3] M. Becker. (2020). *The Mental Health Crisis on Campus and How Colleges Can Fix It*. [Online]. Available: <https://theconversation.com/the-mental-health-crisis-on-campus-and-how-colleges-canfix-it-127875>
- [4] R. P. Auerbach, P. Mortier, R. Bruffaerts, J. Alonso, C. Benjet, P. Cuijpers, K. Demuyttenaere, D. D. Ebert, J. Green, P. Hasking, E. Murray, M. K. Nock, A. D. P. Mak, N. A. Sampson, D. J. Stein, G. Vilagut, A. M. Zaslavsky, and R. C. Kessler, "WHO world mental health surveys international college student project: Prevalence and distribution of mental disorders," *J. abnormal Psychol.*, vol. 127, no. 7, pp. 623–638, 2018.
- [5] *Global Accelerated Action for the Health of Adolescents (AA-HA!): Guidance To Support Country Implementation*, World Health Org., Geneva, Switzerland, 2023.
- [6] A. Duffy, "University student mental health: An important window of opportunity for prevention and early intervention," *Canadian J. Psychiatry*, vol. 68, no. 7, pp. 495–498, 2023.
- [7] R. Gedda-Muñoz, Á. F. Campos, A. V. Sakuda, I. R. Torres, M. C. Fuentes, G. Badicu, T. Herrera-Valenzuela, and P. Valdés-Badilla, "Factors associated with anxiety, depression, and stress levels in high school students," *Eur. J. Invest. Health, Psychol. Educ.*, vol. 13, no. 9, pp. 1776–1786, Sep. 2023.
- [8] A. Gulliver, K. M. Griffiths, and H. Christensen, "Perceived barriers and facilitators to mental health help-seeking in young people: A systematic review," *BMC psychiatry*, vol. 10, no. 1, p. 113, 2010.
- [9] S. Graham, C. Depp, E. E. Lee, C. Nebeker, X. Tu, H.-C. Kim, and D. V. Jeste, "Artificial intelligence for mental health and mental illnesses: An overview," *Current Psychiatry Rep.*, vol. 21, no. 11, pp. 1–18, Nov. 2019.
- [10] H. Abdul Rahman, M. Kwicklis, M. Ottom, A. Amornsriwatanakul, K. H. Abdul-Mumin, M. Rosenberg, and I. D. Dinov, "Machine learning-based prediction of mental well-being using health behavior data from university students," *Bioengineering*, vol. 10, no. 5, p. 575, May 2023.
- [11] S. K. Sharma, A. I. Alutaibi, A. R. Khan, G. G. Tejani, F. Ahmad, and S. J. Mousavirad, "Early detection of mental health disorders using machine learning models using behavioral and voice data analysis," *Sci. Rep.*, vol. 15, no. 1, pp. 1–19, May 2025.
- [12] R. Guidotti, A. Monreale, and D. Pedreschi, "The AI black box explanation problem," *Ercim News*, vol. 2019, pp. 12–13, Jan. 2019.
- [13] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Mach. Intell.*, vol. 1, no. 5, pp. 206–215, May 2019.
- [14] G. Montavon, W. Samek, and K.-R. Müller, "Methods for interpreting and understanding deep neural networks," *Digit. Signal Process.*, vol. 73, pp. 1–15, Feb. 2018.
- [15] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should i trust you?'" in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. discovery data mining*, 2016, pp. 1135–1144.
- [16] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," 2017, *arXiv:1705.07874*.
- [17] L. C. Nnadi, Y. Watanobe, M. M. Rahman, and A. Macgregor John-Otumu, "Prediction of students' adaptability using explainable ai in educational machine learning models," *Appl. Sci.*, vol. 14, no. 12, p. 5141, 2024.
- [18] L. C. Nnadi and Y. Watanobe, "Hierarchical aggregation of local explanations for student adaptability," *Appl. Sci.*, vol. 16, no. 1, p. 333, Dec. 2025.
- [19] M. Christoph, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. Victoria, BC, Canada: Leanpub, 2020.

- [20] X. Ochoa, "Adaptive multilevel clustering model for the prediction of academic risk," in *Proc. Latin Amer. Conf. Learn. Objects Technol. (LACLO)*, Oct. 2016, pp. 1–8.
- [21] S. Wachter, B. Mittelstadt, and C. Russell, "Counterfactual explanations without opening the black box: Automated decisions and the GDPR," *SSRN Electron. J.*, p. 841, May 2018.
- [22] M. Drira, S. B. Hassine, M. Zhang, and S. M. Smith, "Machine learning methods in student mental health research: An ethics-centered systematic literature review," *Appl. Sci.*, vol. 14, no. 24, p. 11738, 2024.
- [23] S. Gunasekara and M. Saarela, "Explainability in educational data mining and learning analytics: An umbrella review," in *Proc. 17th Int. Conf. Educ. Data Mining (EDM)*, 2024, pp. 887–892.
- [24] R. Wang, F. Chen, Z. Chen, T. Li, G. Harari, S. Tignor, X. Zhou, D. Ben-Zeev, and A. T. Campbell, "StudentLife: Assessing mental health, academic performance and behavioral trends of college students using smartphones," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput.*, Sep. 2014, pp. 3–14.
- [25] L. C. López Steinmetz, M. Sison, R. Zhumagambetov, J. C. Godoy, and S. Haufe, "Machine learning models predict the emergence of depression in argentinean college students during periods of COVID-19 quarantine," *Frontiers Psychiatry*, vol. 15, Apr. 2024, Art. no. 1376784.
- [26] A. Baba and K. Bunji, "Prediction of mental health problem using annual student health survey: Machine learning approach," *JMIR Mental Health*, vol. 10, May 2023, Art. no. e42420.
- [27] B. L. Schaab, P. Ü. Calvetti, S. Hoffmann, G. B. Diaz, M. Rech, S. C. Cazella, A. T. Stein, H. M. T. Barros, P. C. D. Silva, and C. T. Reppold, "How do machine learning models perform in the detection of depression, anxiety, and stress among undergraduate students? A systematic review," *Cadernos de Saúde Pública*, vol. 40, no. 11, 2024, Art. no. e00029323.
- [28] L. Luo, J. Yuan, C. Wu, Y. Wang, R. Zhu, H. Xu, L. Zhang, and Z. Zhang, "Predictors of depression among Chinese college students: A machine learning approach," *BMC Public Health*, vol. 25, no. 1, p. 470, Feb. 2025.
- [29] M. Gil, S.-S. Kim, and E. J. Min, "Machine learning models for predicting risk of depression in Korean college students: Identifying family and individual factors," *Frontiers Public Health*, vol. 10, Nov. 2022, Art. no. 1023010.
- [30] N. Meda, S. Pardini, P. Rigobello, F. Visioli, and C. Novara, "Frequency and machine learning predictors of severe depressive symptoms and suicidal ideation among university students," *Epidemiology Psychiatric Sci.*, vol. 32, Jul. 2023, Art. no. e42.
- [31] C. El Mor, M. Jammal, I. Bou-Hamad, S. Hijazi, D. Ayna, M. Romani, and R. Hoteit, "Predictive machine learning models for assessing Lebanese university students' depression, anxiety, and stress during COVID-19," *J. Primary Care Community Health*, vol. 15, Jan. 2024, Art. no. 21501319241235588.
- [32] M. T. Ribeiro, S. Singh, and C. Guestrin, "Anchors: High-precision model-agnostic explanations," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, 2018, pp. 1527–1535.
- [33] I. Covert, S. Lundberg, and S. Lee, "Understanding global feature contributions with additive importance measures," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 17212–17223.
- [34] H. Nori, S. Jenkins, P. Koch, and R. Caruana, "Interpretml: A unified framework for machine learning interpretability," 2019, *arXiv:1909.09223*.
- [35] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," 2017, *arXiv:1702.08608*.
- [36] D. V. Carvalho, E. M. Pereira, and J. S. Cardoso, "Machine learning interpretability: A survey on methods and metrics," *Electronics*, vol. 8, no. 8, p. 832, Jul. 2019.
- [37] J. Amann, A. Blasimme, E. Vayena, D. Frey, and V. I. Madai, "Explainability for artificial intelligence in healthcare: A multidisciplinary perspective," *BMC Med. Informat. Decis. Making*, vol. 20, no. 1, p. 46, Dec. 2020.
- [38] C. Molnar, "Interpretable machine learning: A guide for making black box models explainable," Lulu.com, Morrisville, NC, USA, Tech. Rep., 2019.
- [39] M. R. Zafar and N. Khan, "Deterministic local interpretable model-agnostic explanations for stable explainability," *Mach. Learn. Knowl. Extraction*, vol. 3, no. 3, pp. 525–541, 2021.
- [40] L. C. Nnadi, D. Ding, D. M. Muepu, M. F. I. Amin, and R. Kabir, "Interpretable machine learning for predicting and explaining code submission outcomes in an online judge system," in *Proc. Int. Conf. Ind.*, 2025, pp. 77–88.
- [41] J. Mitchell, N. McLaughlin, and J. Martinez-Del-Rincon, "Generating sparse explanations for malicious Android opcode sequences using hierarchical LIME," *Comput. Secur.*, vol. 137, Feb. 2024, Art. no. 103637.
- [42] P. Knab, S. Marton, and C. Bartelt, "Beyond pixels: Enhancing lime with hierarchical features and segmentation foundation models," 2024, *arXiv:2403.07733*.
- [43] M. Kearns, S. Neel, A. Roth, and Z. S. Wu, "Preventing fairness gerrymandering: Auditing and learning for subgroup fairness," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 2564–2572.
- [44] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, "Model cards for model reporting," in *Proc. Conf. Fairness, Accountability, Transparency*, Jan. 2019, pp. 220–229.
- [45] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé, and K. Crawford, "Datasheets for datasets," in *Proc. 5th Workshop Fairness*, vol. 64, 2021, pp. 86–92.
- [46] H. Suresh and J. Gutttag, "A framework for understanding sources of harm throughout the machine learning life cycle," in *Proc. Equity Access Algorithms, Mech., Optim.*, Oct. 2021, pp. 1–9.
- [47] R. K. Mothilal, A. Sharma, and C. Tan, "Explaining machine learning classifiers through diverse counterfactual explanations," in *Proc. Conf. Fairness, Accountability, Transparency*, Jan. 2020, pp. 607–617.
- [48] A. H. Karimi, G. Barthe, B. Schölkopf, and I. Valera, "A survey of algorithmic recourse: Definitions, formulations, solutions, and prospects," 2021, *arXiv:2010.04050*.
- [49] M. T. Keane, E. M. Kenny, E. Delaney, and B. Smyth, "If only we had better counterfactual explanations: Five key deficits to rectify in the evaluation of counterfactual xai techniques," 2021, *arXiv:2103.01035*.
- [50] F. Afrin, M. Hamilton, and C. Thevathyan, "Exploring counterfactual explanations for predicting student success," in *Proc. Int. Conf. Comput. Sci.*, Springer, 2023, pp. 413–420.
- [51] M. Çavuş and J. Kuzilek, "An effect analysis of the balancing techniques on the counterfactual explanations of student success prediction models," *J. Meas. Eval. Educ. Psychol.*, vol. 15, pp. 302–317, Jan. 2024.
- [52] H. Löfström, T. Löfström, U. Johansson, and C. Sönströd, "Calibrated explanations: With uncertainty information and counterfactuals," *Expert Syst. Appl.*, vol. 246, Jul. 2024, Art. no. 123154.
- [53] M. Domnich and R. Vicente, "Enhancing counterfactual explanation search with diffusion distance and directional coherence," in *Proc. World Conf. Explainable Artif. Intell.*, Jun. 2024, pp. 60–84.
- [54] R. Wang, Y. Xin, Y. Zhang, F. Perez-Cruz, and M. Raubal, "Counterfactual explanations for deep learning-based traffic forecasting," 2024, *arXiv:2405.00456*.
- [55] A. Shamim. (2025). *Student Depression Dataset*. Accessed: Mar. 13, 2025. [Online]. Available: <https://www.kaggle.com/datasets/hopesb/student-depression-dataset>
- [56] P. Muzumdar, G. P. Basyal, and P. Vyas, "An empirical comparison of machine learning models for student's mental health illness assessment," *Asian J. Comput. Inf. Syst.*, vol. 10, no. 1, pp. 1–10, Feb. 2022.
- [57] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bannetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020.



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